The Missing H1/H2 Comparison Package: An Obstruction and the Correct Common Resolution

Repository comparison note

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Abstract

We test the proposed comparison

$$\mathscr{S}_N^{\mathrm{seed}} \longrightarrow \mathscr{H}_N^{\mathrm{adm}} \longrightarrow \mathcal{C}_N$$

one arrow at a time. The first arrow does not exist for the compactifications currently assigned these names. Already in degree five, the stable zero-root configuration forgets a ratio of branch-value scales that the fully branch-marked admissible target retains. Locally the missing datum is the rational function x^3/y^2, and resolving it requires the normalized blowup of (x^3, y^2). Thus the admissible-cover closure is a modification, not an identified copy, of the root-stable quotient.

We replace the false arrow by the normalization of the graph. The second arrow, from the normal admissible incidence to the repository's normalized finite incidence, is then constructed by the universal property of normalization. We state exact criteria for finiteness, birationality, normalization identification, compatibility with the selected affine root, and agreement of completed local rings. This settles the comparison question negatively as stated and gives the minimal corrected package needed by downstream arguments.

1 The objects and the claimed arrows

Work over an algebraically closed field k of characteristic zero. Put $n = N - 2$. Let

$$\mathscr{S}_N^{\mathrm{seed}} = [\overline{M}_{0,N}/S_{n-1}]$$

be the collision-separating compactification with the simple root 1 selected, and let $\mathscr{S}_N^{\circ}$ be its smooth, distinct-root open. Let $\mathscr{H}_N^{\mathrm{adm}}$ be the normalization of the closure of this open in a separated admissible-cover stack which retains the stabilized target branch divisor. All finite branch points may be unordered; the argument below only uses whether their stabilized target is reducible.

Finally, let B_N be the chosen repository base and let $\mathcal{C}_N$ be the normalization of the scheme-theoretic closure of the relevant generic finite incidence over B_N. Thus $\mathcal{C}_N \to B_N$ is finite and normal and comes with a fixed generic incidence algebra. This definition includes either the residual degree-n root cover or, after replacing the generic algebra, the degree-N inverse incidence. The two degrees must not be mixed.

On $\mathscr{S}_N^{\circ}$ the polynomial and hence its admissible cover are defined. This gives a rational map

$$\phi^{\circ} : \mathscr{S}_N^{\mathrm{seed}} \dashrightarrow \mathscr{H}_N^{\mathrm{adm}}. \tag{1.1}$$

The issue is whether (1.1) is regular at the boundary.

2 A degree-five obstruction to the first arrow

Take $N = 5$ and write the simple zero-fiber roots as $1, a, b$. On the clean open put

$$F_{a,b}(W) = W^2(W-1)(W-a)(W-b), \qquad H_{a,b}(W) = -\frac{F_{a,b}(W)}{F'_{a,b}(1)}. \tag{2.1}$$

Then $H(0) = H'(0) = H(1) = 0$ and $H'(1) = -1$.

Near the codimension-two boundary point at which a collides with 0 and b collides with 1, write

$$a = x, \qquad b = 1 + y. \tag{2.2}$$

The completed local ring of the labelled root-stable space at this point is $k[[x, y]]$. The special stable source is independent of the relative rates of x and y: it has a 0-tail carrying $(0, a)$, a 1-tail carrying $(1, b)$, and a central component carrying ∞ and the two attaching nodes.

There are two moving critical values near the marked target value zero. One comes from the triple cluster $(0, 0, a)$ and the other from the pair $(1, b)$. Denote them by λ_0 and λ_1, respectively.

Lemma 2.1 (branch-scale asymptotics)**.** *In the fraction field of $k[[x, y]]$, along every trait on which $3v(x) > v(y) > 0$, one has*

$$\lambda_0 = -\frac{4}{27}\frac{x^3}{y}(1 + o(1)), \qquad \lambda_1 = -\frac{1}{4}y\,(1 + o(1)). \tag{2.3}$$

Consequently

$$\frac{\lambda_0}{\lambda_1} = \frac{16}{27}\frac{x^3}{y^2}(1 + o(1)). \tag{2.4}$$

Proof. Since $F'_{a,b}(1) = (1-a)(1-b) = -(1-x)y$, equation (2.1) gives

$$H_{a,b}(W) = \frac{F_{a,b}(W)}{(1-x)y}. \tag{2.5}$$

Put $W = xX$. Uniformly along the stated traits,

$$H_{a,b}(xX) = \frac{x^3}{y}X^2(X-1)(1 + o(1)).$$

Besides the fixed critical point $X = 0$, the moving critical point tends to $X = 2/3$, giving the first formula in (2.3). Put instead $W = 1 + yX$. Then

$$H_{a,b}(1 + yX) = yX(X-1)(1 + o(1)),$$

whose moving critical point tends to $X = 1/2$. This gives the second formula and hence (2.4). □

Theorem 2.2 (nonextension)**.** *For $N \geq 5$, the rational map (1.1) does not extend to a morphism from the root-stable quotient to any separated admissible-cover compactification that retains the stabilized target branch divisor.*

Proof. It is enough to work on the labelled degree-five chart; fixed spectator roots give the assertion in every higher degree. Consider two traits through the same point $x = y = 0$:

$$\begin{array}{c|cc|c} & x & y & \lambda_0/\lambda_1 \\ \hline T_A & t & t & (16/27)t(1+o(1)) \\ T_B & t^2 & t^3 & (16/27)(1+o(1)). \end{array} \tag{2.6}$$

Both have the same stable marked source described after (2.2). On the target, retain the four points $0, \lambda_0, \lambda_1, \infty$ and forget all other branch points. Along T_A their stable four-pointed limit lies on the boundary of $\overline{M}_{0,4}$ because $\lambda_0/\lambda_1 \to 0$. Along T_B the limit is smooth because the ratio tends to $16/27 \notin \{0, 1, \infty\}$. The same distinction survives if λ_0 and λ_1 are unordered.

Thus the two admissible covers have nonisomorphic stabilized targets. If a morphism (1.1) existed, both traits would send their common closed source point to one point of the admissible-cover stack. Separatedness and stable reduction force that point to be each of the two nonisomorphic limits, a contradiction. □

Corollary 2.3 (the missing local modification)**.** *At the boundary point above, the rational target-stabilization map contains the rational map*

$$\operatorname{Spec} k[[x, y]] \dashrightarrow \mathbb{P}^1, \qquad [x^3 : y^2]. \tag{2.7}$$

Its graph is the blowup of (x^3, y^2), and its normal graph is the normalized blowup of this monomial ideal. In particular the graph projection is not finite and the completed local ring is not the original $k[[x, y]]$.

Proof. Equation (2.4) identifies the target cross-ratio up to a unit with x^3/y^2. The closure of the graph of a rational map $[f : g]$ is the blowup of (f, g). Its fiber over $(x, y) = (0, 0)$ contains the exceptional projective curve; normalization does not make that fiber finite. □

This is the precise point missed by a proof based only on DVR extension. Every individual trait has an admissible stable limit, but those limits do not descend to a morphism from the unmodified two-dimensional base.

3 The corrected first comparison

Let

$$\Gamma_N = \operatorname{Norm} \overline{\Gamma_{\phi^\circ}} \subset \mathscr{S}_N^{\mathrm{seed}} \times \mathscr{H}_N^{\mathrm{adm}} \tag{3.1}$$

be the normalization of the graph closure, interpreted on labelled atlases and descended equivariantly to the quotient stacks. It comes with

$$\mathscr{S}_N^{\mathrm{seed}} \xleftarrow{p} \Gamma_N \xrightarrow{q} \mathscr{H}_N^{\mathrm{adm}}. \tag{3.2}$$

Proposition 3.1 (common-resolution package)**.** *The maps in (3.2) are proper and birational onto the components containing $\mathscr{S}_N^\circ$. The following are equivalent:*

1. *ϕ° extends to $\mathscr{S}_N^{\mathrm{seed}} \to \mathscr{H}_N^{\mathrm{adm}}$;*
2. *p is an isomorphism;*
3. *p is finite.*

For $N \geq 5$ these conditions fail for the fully branch-marked admissible compactification.

Proof. Properness follows from graph closure in the product of proper stacks, and birationality follows because both projections restrict to the identity on the common dense open. An extension gives a section of p whose graph is all of Γ_N; separatedness gives uniqueness, hence p is an isomorphism. Conversely an inverse to p gives the extension. If p is finite, it is finite birational onto the normal stack $\mathscr{S}_N^{\mathrm{seed}}$, hence is an isomorphism. Theorem 2.2 and Corollary 2.3 give the final assertion. □

Thus the corrected first arrow is not $\mathscr{S}_N^{\mathrm{seed}} \to \mathscr{H}_N^{\mathrm{adm}}$ but the correspondence (3.2). Equivalently, one may replace the source by the logarithmic/toroidal modification Γ_N that records relative branch-value scales. In the local chart (2.2), the first required subdivision is the normalized blowup of (x^3, y^2).

4 Construction of the incidence contraction

The second comparison arrow is formal once the generic algebra and the base are fixed correctly.

Proposition 4.1 (normalization contraction)**.** *Let B be an integral locally Noetherian base, let $K/k(B)$ be the fixed finite generic incidence algebra (a field component, or a specified product treated componentwise), and let*

$$C = \mathrm{Norm}_B(K)$$

be finite over B. Let H be a normal integral algebraic stack, dominant and proper over B, whose generic function field is the chosen component of K. Then the generic comparison extends uniquely to a B-morphism

$$c : H \longrightarrow C. \tag{4.1}$$

It is proper and birational. It is finite if and only if it is quasi-finite, and in that case it is an isomorphism.

Proof. The assertion is smooth-local on H and affine-local on B. If $B = \operatorname{Spec} A$ and $C = \operatorname{Spec} \overline{A}$, every element of $\overline{A}$ is integral over A. Its image in the common function field is therefore integral over every normal local ring of H. It belongs to that local ring, so the generic map extends and the local extensions glue uniquely. This is the universal property of normalization. Properness follows because H is proper over B and C is separated over B. The generic comparison makes c birational. A proper quasi-finite morphism is finite; a finite birational morphism to normal C is an isomorphism. □

Apply Proposition 4.1 with $H = \mathscr{H}_N^{\mathrm{adm}}$ and $C = \mathcal{C}_N$, separately for:

1. the residual degree-n zero-fiber algebra after removing the fixed length-two summand; and
2. the degree-N inverse-incidence algebra $k(s,t)[W]/(H(W) - sW + t)$.

This constructs the arrow

$$\mathscr{H}_N^{\mathrm{adm}} \longrightarrow \mathcal{C}_N \tag{4.2}$$

provided the admissible object in (4.2) is the corresponding marked-point incidence and its generic algebra has first been verified component by component. Normalized Stein factorization is an equivalent construction: the normal H maps to the normalization of $\mathrm{Spec}_B(f_*\mathcal{O}_H)$ in the fixed generic algebra.

5 Compatibility with the selected affine root

On the clean open, the chosen point p_1 maps under (4.2) to the tautological root of the finite incidence. Both $\mathcal{C}_N \to B_N$ and the coarse marked root cover are separated. Therefore equality on the dense open implies equality after pullback to Γ_N wherever the two sections extend. The finite-cover DVR lemma then gives, for each fixed generic selected root, a unique extension over every trait.

This proves traitwise compatibility but not descent from Γ_N to $\mathscr{S}_N^{\mathrm{seed}}$. Descent would require the selected section to be constant on every fiber of p in (3.2). The abstract selected-root cover does satisfy that statement, but the full admissible target does not: Corollary 2.3 exhibits its varying exceptional parameter.

6 Completed local rings and the exact final status

Let $\gamma \in \Gamma_N$ map to $h \in \mathscr{H}_N^{\mathrm{adm}}$ and $z \in \mathcal{C}_N$. The morphism (4.2) induces

$$\widehat{\mathcal{O}}_{\mathcal{C}_N,z} \longrightarrow \widehat{\mathcal{O}}_{\mathscr{H}_N^{\mathrm{adm}},h}. \tag{6.1}$$

An isomorphism in (6.1) is a consequence of an actual global morphism plus a formal calculation; it cannot be used to manufacture that morphism. For a proper birational map of excellent normal spaces, isomorphism of these completed maps at every point implies that the map is formally quasi-finite, hence quasi-finite, finite, and finally an isomorphism.

At a pure root collision with all target-scale data fixed, the repository's invariant calculation

$$k[[e_1,\ldots,e_\mu,T]]/(T^\mu - e_1T^{\mu-1} + \cdots + (-1)^\mu e_\mu) \tag{6.2}$$

is the correct completed finite root cover. It proves agreement of the root-selection factor. It does not include the target cross-ratio x^3/y^2 in (2.4). At the simultaneous collision of Section 2 the source completion is $k[[x,y]]$, whereas the graph/admissible side contains the normalized blowup chart of (x^3,y^2). Hence the full completed local rings do not agree.

The requested checklist therefore has the following unconditional answer:

boundary extension	yes after replacing the first arrow by Γ_N;
normalization compatibility	yes, by Proposition 4.1;
selected affine root	yes on Γ_N and on every trait;
finiteness of $\Gamma_N \to \mathscr{S}_N^{\mathrm{seed}}$	no for $N \geq 5$;
birationality onto the claimed components	yes;
normality/normalization identification	yes for $\mathcal{C}_N$ by definition;
full completed-ring agreement	no at the two-scale boundary;
root-factor completed-ring agreement	yes, by (6.2).

Accordingly, H1/H2 cannot be repaired by proving the originally displayed arrows. The correct global package is

$$\boxed{\mathscr{S}_N^{\mathrm{seed}} \xleftarrow{\text{proper birational}} \Gamma_N \xrightarrow{\text{proper birational}} \mathscr{H}_N^{\mathrm{adm}} \xrightarrow{\text{normalization contraction}} \mathcal{C}_N.}$$

Downstream statements using only the finite selected-root cover and its DVR extension survive. Statements identifying the entire root-stable and branch-stable compactifications must either use Γ_N or forget enough target branch-scale data for the obstruction to disappear.